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Roll No

EC-605 (GS)
B.E. VI Semester Examination, June 2020
Grading System (GS)
VLSI Circuits and Systems
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. Give a brief introduction on Controllers used in MOS devices. Give a classification of them and explain each one of them with the help of suitable example.
2. A sequential circuit has 2D ff's A and B an input x and output Y is specified by the following next state and output equations.
 $A(t+1) = Ax + Bx$
 $B(t+1) = A'x$
 $Y = (A+B)x'$
Derive the state table and the state diagram.
3. Design a Binary to Gray code converter using
i) PROM ii) PLA iii) PAL
4. Explain the following memory elements.
i) D Flip-Flop
ii) T Flip-Flop
iii) S-R Flip-Flop
iv) JK Flip-Flop
5. What is system controller? Discuss the design aspect of controller phases. Design a controller for binary multiplier.
6. Describe the following with the help of CMOS logic.
i) Inverter
ii) Compound gates
iii) Multiplexers
7. State the difference between Mealy and Moore machine. Given an example and draw the state transition diagram.
8. Briefly explain the following.
a) Inflammation Loss-less machines
b) FPGA technologies
c) Test Vector
